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14. LOG ANALYSIS FOR DETECTION AND RESPONSE

AIM

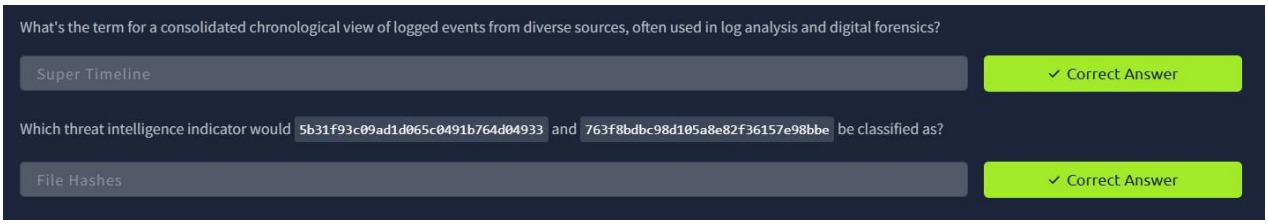
To understand the fundamentals of log analysis, its role in incident detection and response, and the tools used for analyzing logs in the TryHackMe room "Intro to Log Analysis".

PROCEDURE

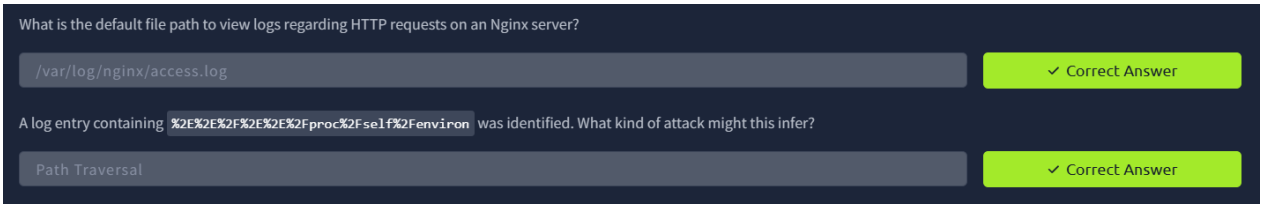
1. **Access the "Intro to Log Analysis" room in TryHackMe and study the basics of logging and log analysis techniques.**
2. **Learn how system, network, and application logs are used to identify suspicious or malicious activities.**
3. **Explore commonly used log analysis tools such as grep, tail, wc, CyberChef, and regular expressions (regex).**
4. **Complete the investigation tasks and answer the questions provided in the room.**

QUESTIONS

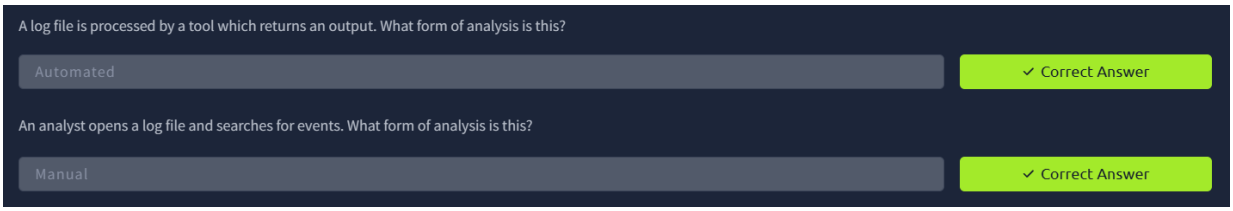
TASK 3 – Investigation Theory



TASK 4 – Detection Engineering



TASK 5 – Automated vs. Manual Analysis



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TASK 6 – Log Analysis tools: Command Linux

Use `cut` on the `apache.log` file to return only the URLs. What is the flag that is returned in one of the unique entries?

c701d43cc5a3acb9b5b04db7f1be94f6 — ✓ Correct Answer

In the `apache.log` file, how many total HTTP 200 responses were logged?

52 — ✓ Correct Answer

In the `apache.log` file, which IP address generated the most traffic?

145.76.33.201 — ✓ Correct Answer

What is the complete timestamp of the entry where `110.122.65.76` accessed `/login.php`?

31/Jul/2023:12:34:40 +0000 — ✓ Correct Answer

TASK 7 – Log Analysis Tools: Regular Expression

How would you modify the original `grep` pattern above to match blog posts with an ID between 20-29?

post=2[0-9] — ✓ Correct Answer

What is the name of the filter plugin used in Logstash to parse unstructured log data?

Grok — ✓ Correct Answer

TASK 8 – Log Analysis Tools: Cyberchef

Locate the "loganalysis.zip" file under `/root/Rooms/introloganalysis/task8` and extract the contents.

No answer needed — ✓ Correct Answer

Upload the log file named "access.log" to CyberChef. Use regex to list all of the IP addresses. What is the full IP address beginning in 212?

212.14.17.145 — ✓ Correct Answer

Using the same log file from Question #2, a request was made that is encoded in base64. What is the decoded value?

THM{CYBERCHEF_WIZARD} — ✓ Correct Answer

Using CyberChef, decode the file named "encodedflag.txt" and use regex to extract by MAC address. What is the extracted value?

08-2E-9A-4B-7F-61 — ✓ Correct Answer

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TASK 9 – Log Analysis Tools: Yara and Sigma

RESULT

Successfully learned the importance of log analysis, best practices, and tools used for detecting, investigating, and responding to security incidents.